

NUKUS, Uzbekistan

WMO: 382640

Lat: 42.483N

Lon: 59.617E

Elev: 253

StdP: 14.56

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	2.9	-10.1	3.2	1.3	-4.4	4.4	7.0	22.8	32.4	20.4	30.7	7.0	10	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	103.7	70.9	100.7	69.6	98.4	69.1	76.3	95.8	74.3	93.4	72.4	91.5	10.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.2	112.1	87.7	67.9	103.3	85.9	65.7	95.6	84.4	40.0	95.8	38.0	93.0	36.2	91.3	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	20.1	17.7	-6.1	108.4	7.7	2.7	-11.6	110.4	-16.1	112.0	-20.4	113.5	-26.0	115.5
			-6.6	77.9	7.5	3.9	-12.0	80.7	-16.4	83.0	-20.6	85.1	-26.1	88.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	25.3	29.8	44.4	59.0	71.9	80.7	84.0	80.5	68.3	54.3	39.5	29.1
	DBStd	22.49	11.32	11.29	10.07	9.39	6.85	5.79	5.18	6.12	7.18	8.88	9.75	9.66
	HDD50	2631	766	568	230	27	0	0	0	0	1	58	332	649
	HDD65	5369	1231	985	639	224	21	1	0	1	46	344	764	1113
	CDD50	4717	0	3	58	298	679	922	1053	945	549	192	18	0
	CDD65	1980	0	0	2	45	234	473	588	480	145	13	0	0
	CDH74	27651	0	1	41	580	2953	6708	8645	6634	1880	207	2	0
	CDH80	15462	0	0	7	197	1356	3905	5275	3895	789	38	0	0
Wind	WSAvg	8.3	7.9	8.9	9.6	9.6	8.8	8.8	8.5	8.5	7.6	7.0	7.1	7.6
Precipitation	PrecAvg	4.2	0.4	0.3	0.6	0.7	0.6	0.1	0.1	0.0	0.1	0.2	0.3	0.5
	PrecMax	9.4	0.9	0.9	2.2	3.1	2.1	1.1	0.7	0.6	0.6	1.2	2.0	1.2
	PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.0	0.3	0.3	0.6	0.7	0.6	0.2	0.2	0.1	0.2	0.3	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	62.9	80.0	91.1	98.6	105.6	108.0	106.7	98.7	84.2	70.0	55.6
		MCWB	43.5	49.6	57.5	64.2	65.9	69.8	70.3	74.1	69.6	61.1	54.7	45.8
	2%	DB	46.2	54.4	71.4	86.0	94.5	102.0	104.2	102.2	91.8	79.7	63.0	49.6
		MCWB	41.0	44.6	53.9	60.2	63.7	69.3	71.2	71.0	66.1	58.0	50.3	43.2
	5%	DB	42.1	48.6	65.8	80.7	91.2	98.8	101.0	99.0	88.1	75.2	58.9	45.0
		MCWB	38.2	40.8	51.1	58.4	63.3	68.3	70.1	69.8	63.5	56.2	48.2	40.1
	10%	DB	38.6	44.0	60.6	75.9	87.8	96.2	98.5	96.4	84.3	70.9	53.9	41.3
		MCWB	35.7	38.3	48.3	56.6	62.0	67.9	69.7	68.9	61.7	54.3	45.5	37.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	49.7	58.9	66.7	71.3	77.3	79.5	78.6	72.7	64.1	57.3	47.2
		MCDB	49.0	60.7	75.1	85.1	90.2	96.3	97.6	100.0	91.5	79.8	67.3	53.2
	2%	WB	41.2	45.4	55.2	63.1	67.9	74.7	76.7	75.8	69.1	60.0	52.7	43.6
		MCDB	45.5	52.7	68.7	80.2	86.3	93.2	95.6	96.7	86.6	75.1	61.4	48.6
	5%	WB	38.4	41.5	52.1	60.6	65.7	72.4	74.5	73.5	66.5	57.6	49.4	40.5
		MCDB	41.6	47.3	63.7	76.2	83.5	91.5	93.1	93.5	83.8	71.8	56.1	44.3
	10%	WB	35.8	38.5	49.4	58.1	63.9	70.4	72.6	71.2	63.8	55.3	46.3	37.8
		MCDB	38.2	43.2	59.4	72.9	82.2	89.4	91.3	90.6	80.2	68.0	52.7	41.2
Mean Daily Temperature Range	5% DB	MDBR	14.1	16.9	19.8	22.0	24.2	25.0	24.5	26.2	26.5	24.1	18.5	14.6
		MCDBR	16.6	23.0	26.7	27.8	29.1	29.0	28.1	29.6	30.6	29.7	25.6	19.6
		MCWBR	11.7	13.8	14.1	12.2	11.2	10.6	10.1	11.0	13.8	15.0	14.8	13.8
	5% WB	MCDBR	15.3	20.8	24.3	24.1	23.9	24.7	24.3	26.6	27.2	25.5	20.8	17.0
		MCWBR	11.3	13.3	13.9	12.4	10.8	10.8	10.7	11.6	14.0	14.1	12.9	12.7
Clear-Sky Solar Irradiance	taub	0.383	0.413	0.474	0.530	0.536	0.525	0.509	0.481	0.450	0.456	0.418	0.386	
	taud	2.049	1.988	1.895	1.800	1.829	1.880	1.943	1.989	2.015	1.954	2.016	2.047	
	Ebn at Noon	224	239	239	237	239	243	245	247	245	220	208	208	
	Edh at Noon	36	44	55	65	65	62	58	54	48	45	37	33	
All-Sky Solar Radiation	RadAvg	500	824	1267	1712	2117	2381	2306	2127	1684	1120	674	458	
	RadStd	62	89	106	114	119	104	75	63	65	66	80	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.24	-4.03	-5.02	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	+1.49	N/A	-1.21	N/A	N/A	N/A	+94

Nomenclature: See separate page